Multi-View Entity Resolution

Three Representation Levels and Three Fusion Strategies

Xinmin Chu   David Degras-Valabregue

Department of Mathematics, UMass Boston

May 20, 2026

Entity Resolution Is a Multi-Field Problem

Entity resolution (ER) identifies records referring to the same real-world entity across one or more datasets.

Standard pipeline:

1. **Block** — reduce $O(n^2)$ pairs to a candidate set
2. **Compare** — compute field similarities for each pair
3. **Classify** — decide match / non-match per pair
4. **Cluster** — group matched records into entity clusters

Two open questions for multi-field ER:

- At what *level* should field values be represented?
- *When* should multi-field information be combined?

Field	Type
Last name	short text
First name	short text
DOB	date / numeric
Street	medium text
City	categorical
Title	long text
Authors	list
Year	numeric

Three Levels of Representation

The representation level determines which similarity methods apply. **This paper focuses on Level 3**, with Levels 1 and 2 as baselines (Exp 5).

Level 1 — Embedding

- Field value → dense $\mathbb{R}^d$ via LLM / BERT
- Similarity: cosine in embedding space
- ✓ Semantic richness; handles typos
- ✓ No feature engineering needed
- × Black-box; expensive; needs GPU

Level 2 — Bag of Words

- Field value → sparse token vector (n-grams, IDF)
- Similarity: Jaccard / TF-IDF cosine
- ✓ Interpretable; fast; no training
- ✓ Works on any text field
- × Order-blind; no semantics

Level 3 — Field Similarity

- Score $S_{ij}^{(k)}$ via Jaro-Winkler, exact match, date diff
- Similarity: engineered per field type
- ✓ Domain-precise; fully interpretable
- ✓ NA-aware; sparse-matrix-friendly
- × Requires manual design per field

Within Level 3, the key question is *when and how* to fuse the p similarity matrices $S^{(1)}, \ldots, S^{(p)}$ — which is where the three fusion strategies (A, B, C) come in.

The Complete ER Pipeline: Classify Before Cluster

Key principle: similarity scores feed an explicit *binary classifier* before clustering. Never cluster on raw similarity scores directly.

Block → Represent (Level 1/2/3) → Compare $S^{(k)}$ → Fuse (A, C; skip B) → **Classify** match/non-match → Cluster (CC, etc.)

Why the explicit classification step:

- Fellegi–Sunter classifies each pair as M, U, or P before any grouping — classification *is* canonical ER
- Binary match graph $\Rightarrow$ transitive closure via connected components; raw S_{ij} does not support this cleanly
- Decision threshold is a principled boundary, not a latent clustering parameter

NA-Aware Combination (Level 3):

$$S(i,j) = \frac{\sum_{k:\, S_{ij}^{(k)} \neq \mathtt{NA}} w_k S_{ij}^{(k)}}{\sum_{k:\, S_{ij}^{(k)} \neq \mathtt{NA}} w_k}$$

Never impute. Adaptive denominator handles missingness.

Taxonomy: Nine Ways to Fuse $S^{(1)}, \ldots, S^{(p)}$

#	Method	Strategy	Role in this paper
1	Simple weighted aggregation	A — Late clustering	Primary Paradigm A baseline
2	Multi-view spectral clustering	A — Late clustering	Paradigm A variant
4	Multiple Kernel Learning (MKL)	A — Late clustering	Theoretical frame for A
7	Tensor factorization (PARAFAC)	A — Late clustering	Related work
6	Ensemble / consensus clustering	B — Early clustering	Primary Paradigm B
8	Probabilistic LVM (Bayesian)	B — Early clustering	Related work
3	Similarity Network Fusion (SNF)	C — Iterative/hybrid	Paradigm C comparator
5	Multiplex community detection	C — Iterative/hybrid	Paradigm C comparator
9	Deep learning / GNN	C — Iterative/hybrid	Bridges Levels 1 and 3

A: normalize → fuse matrices → classify → cluster once

B: classify per field → combine binary decisions

C: iterative or joint optimization across views

ERBOT now: #1, 4, 6 fully; #5, 9 partially

Add for paper: #2 (spectral), #3 (SNF), #5 (multiplex)

Related work only: #7 (tensor), #8 (Bayesian)

Related Work

Strategy A — Late Clustering

- Fellegi & Sunter (1969) — field-independent likelihood ratios
- Lanckriet et al. (2004) — MKL: $K = \sum_k w_k K_k$
- Kumar et al. (2011) — co-regularized multi-view spectral

Strategy B — Early Clustering

- Strehl & Ghosh (2002) — CSPA / MCLA / HGPA
- Fred & Jain (2005) — Evidence Accumulation (EAC)

Strategy C — Hybrid

- Wang et al. (2014) — SNF: iterative graph diffusion
- Mucha et al. (2010) — multiplex community detection

ER & Multi-view Foundations

- Jaro (1989); Winkler (1990) — string similarity
- Christen (2012) — block–compare–classify
- Xu et al. (2013) — early/late fusion (multi-view survey)

Deep / Embedding ER

- Mudgal et al. (2018) — DeepMatcher
- Li et al. (2020) — Ditto (BERT)

Conceptual regimes:

A $=$ noisy variants of one signal $\rightarrow$ fuse first
B $=$ complementary notions $\rightarrow$ multi-view
C $=$ different relation types $\rightarrow$ multilayer

Gap: No prior ER work compares all three strategies across field types, scales, and missingness.

$$\{S^{(k)}\} \xrightarrow{\text{normalize, fuse}} S = \sum_{k=1}^{p} w_k S^{(k)} \xrightarrow{\text{classify}} M_{ij} \in \{0,1\} \xrightarrow{\text{cluster}} \hat{\mathcal{C}}_A$$

Weight learning (`er_weights()`):

Method	When to use	Connection to MKL
Equal ($w_k = 1/p$)	No prior knowledge	MKL uniform
Fellegi–Sunter EM	Unsupervised	Unsupervised MKL
ARI-based	Ground truth available	Supervised MKL
Bimodality / variance	Field discriminability	Heuristic MKL

Paradigm A $\equiv$ MKL. When $S^{(k)}$ are PSD kernels, $S = \sum_k w_k S^{(k)}$ is the standard MKL combination (Lanckriet 2004; Gönen & Alpaydın 2011). Variants: Method 2 (average Laplacians $L = \sum_k w_k L_k$, spectral); Method 4 (kernel k-means on $K = \sum_k \beta_k K_k$).

Wins when: fields are complementary noisy variants of the same signal; scales are comparable; ground truth is available.

Strategy B: Early Clustering

$$S^{(k)} \xrightarrow{\text{classify}} M_{ij}^{(k)} \in \{0,1\} \xrightarrow{\text{cluster}} \hat{\mathcal{C}}^{(k)},\ k = 1,\ldots,p \xrightarrow{\text{consensus}(\alpha)} \hat{\mathcal{C}}_B$$

Consensus — adaptive majority vote (p_{ij} = non-missing fields for pair (i,j)):

$$r_i \sim r_j \text{ in } \hat{\mathcal{C}}_B \iff \tfrac{1}{p_{ij}}\textstyle\sum_{k:\, S_{ij}^{(k)} \neq \mathtt{NA}} M_{ij}^{(k)} \geq \alpha$$

Advantages

- Per-field classification is independent ⇒ parallelizable
- One bad field dilutes, does not dominate
- No cross-field weight learning; no joint normalization
- α directly interpretable: fraction of views agreeing

Limitations

- Loses all inter-field correlations
- Each field must be individually classifiable
- Consensus fragments when fields strongly conflict
- Missing field shrinks effective p; adjust denominator

Wins when: one or few fields are individually sufficient; fields are incommensurable or noisy; n is large; missingness is high.

Strategy C: Iterative / Hybrid Fusion

Method 3 — Similarity Network Fusion (SNF)

$S^{(k)} \to$ row-stochastic $P^{(k)}$. Iterate until convergence:

$$P^{(k)} \leftarrow P^{(k)} \cdot \tfrac{1}{m-1} \textstyle\sum_{l \neq k} P^{(l)} \cdot \left(P^{(k)}\right)^{\top}$$

Cluster the fused graph spectrally.

Best for: heterogeneous complementary fields, noisy modalities. (Wang et al. 2014)

Method 5 — Multiplex Community Detection

Each $S^{(k)}$ is one *layer* of a multiplex graph. Maximize modularity jointly:

$$Q = \sum_{k,ij} \left[A_{ij}^{(k)} - \gamma_k P_{ij}^{(k)}\right] \delta(c_i, c_j)$$

Leiden/Louvain with inter-layer coupling.

Best for: fields representing *different relation types*. (Mucha et al. 2010)

Method 9 — Deep / GNN (bridge to Level 1)

GNNs take $\{S^{(k)}\}$ as graphs, learn a joint embedding via message-passing, then classify pairs. Bridges Level 3 inputs with Level 1 outputs. Best for large-scale nonlinear data; less interpretable.

Normalization: Often More Important Than the Algorithm

Different fields have incompatible scales. Strategy A is directly distorted by scale mismatch. B and C are more robust but not immune.

Method	**When to use in ER**
Row-stochastic	Sparse matrices; required by SNF
Min-max	Bounded $[0, 1]$ similarities
z-score	Wildly different distributions
Graph Laplacian	Spectral clustering (Method 2)

Field-specific notes:

- Jaro-Winkler $\in [0, 1]$: min-max stable
- Date / numeric: unbounded — z-score or clip
- TF-IDF cosine: already normalized
- SNF: row-stochastizes $S^{(k)}$ internally

Exp 1 treats normalization as a crossed factor.

Key finding from 9-method review

Normalization is often more important than the clustering algorithm. Skipping it is the most common reason Strategy A underperforms on heterogeneous-field datasets.

Research Questions

RQ1 — Strategy Accuracy

Does Strategy A, B, or C recover entities more accurately across datasets and field configurations?

RQ2 — Field Quality and Missingness

How do field discriminability and missingness rate affect each strategy? Which degrades most gracefully?

RQ3 — Scalability

How does each strategy scale with n? Does parallel per-field classification give Strategy B an advantage at large scale?

RQ4 — Normalization Sensitivity

Does normalization strategy interact with fusion strategy? Does Strategy A degrade more under scale mismatch than B or C?

RQ5 — Representation Level

How does Level 1 (embedding), Level 2 (BoW), or Level 3 (field similarity) interact with fusion strategy?

RQ6 — Predictability

Can field-level diagnostics (cardinality, missingness, per-field ARI) predict the winning strategy before experiments run?

Benchmark Datasets

All four datasets have genuine multi-field structure.

Dataset	Records	Entities	Fields	Mode	Primary role
CORA	1,879	182	15	Dedup	Multi-field ablation (RQ1, RQ2)
Restaurant	864	754	5	Linkage	Clean small benchmark
DBLP-ACM	4,910	2,690	4	Linkage	Missingness robustness (RQ2, RQ4)
NCVR	100,000	98,521	4	Linkage	Large-scale stress test (RQ3)

CORA — 15 fields: long text (title, abstract), lists (authors), categoricals (venue). Rich field set; ideal for ablation.

Restaurant — name, address, city, phone, cuisine. Fodors vs. Zagats; all three representation levels applicable.

DBLP-ACM — title, authors, venue, year. Moderate size; used for controlled missingness injection (Exp 4).

NCVR — givenname, surname, suburb, postcode. 100K records (10 parties $\times$ 10K); scalability benchmark (Exp 3).

Working Hypotheses

Strategy A wins when...

- Fields are complementary: each weak alone, strong together
- Scales are comparable (or normalization corrects them)
- Ground truth available for supervised weights

Strategy B wins when...

- One or few fields are individually sufficient
- Fields are incommensurable or noisy
- n is large; missingness is high

Strategy C wins when...

- Views capture complementary local structure (SNF)
- Fields represent different relation types (multiplex)
- Nonlinear cross-field interactions present (GNN)

Central claim

The winning strategy is predictable from field-level diagnostics (cardinality, missingness, per-field ARI) before any experiments run.

Experiment Design

Exp 1 — Main accuracy comparison (all 4 datasets, 5-fold CV)

Factor	**Levels**
Strategy	A (late clustering), B (early clustering), C (SNF, multiplex)
Weight / α	Equal, Fellegi–Sunter EM; $\alpha \in \{0.3, 0.5, 0.7\}$
Normalization	None, row-stochastic, z-score, min-max
Clustering	Connected components (primary); Louvain, HC-Average (comparison)
Metric	ARI (primary); B-cubed P, R, F_1; Pairwise P, R, F_1

Exp 2 — Field ablation (CORA): remove one field at a time; compare ARI drop across A, B, C.

Exp 3 — Scalability (NCVR, $n \in \{1\text{K}, 10\text{K}, 50\text{K}, 100\text{K}\}$): wall-clock time and peak memory per strategy.

Exp 4 — Missingness robustness (DBLP-ACM): mask fields at 0%–50%; compare ARI, B-cubed F_1, pairwise P/R across A, B, C.

Exp 5 — Representation level (Restaurant, DBLP-ACM): Level 1 (BERT) vs. Level 2 (TF-IDF BoW) vs. Level 3 (field similarity), Strategy A fixed.

Method Selection Guide

The empirical claim experiments will validate; output of `er_compare_paradigms()`.

ER Situation	Recommended	Rationale
Fields are noisy variants of entity identity	Strategy A	Fusion averages noise across views
Fields capture orthogonal dimensions	Strategy B	Independence of views is a feature
Heterogeneous views, local structure	C — SNF	Iterative diffusion integrates complementary info
Fields represent different relation types	C — Multiplex	Multilayer graph is the right framing
High missingness (>30%)	A (NA-aware) or B	NA formula or field skip both handle absence
$n > 50\text{K}$, scalability critical	Strategy B	Parallel per-field classification
Ground truth available	A (ARI weights)	Supervised MKL optimizes weights

Implementation: erbot R Package

Stage	Strategy A	Strategy B	Strategy C
Represent	`er_embed()` (L1)	`er_bow()` (L2)	`er_similarity()` (L3)
Normalize	`er_normalize(method=)` — row-stoch, *z*-score, min-max		
Fuse	`er_combine()` + `er_weights()`	*(skip)*	`er_snf()`, `er_multiplex()`
Classify	`er_classify(threshold=)` — outputs binary match graph M		
Cluster	`er_cluster()` on M	`er_classify()` $\times p$ → `er_consensus(`α`)`	`er_cluster()` on fused graph
Compare	`er_compare_paradigms()` *(in progress)*		

Implemented: A (✓), B (✓); C — Louvain/Leiden and neural encoder (partial).

To add: `er_snf()`, `er_normalize()`, `er_bow()`, `er_classify()`.

Paper Structure and Target Venue

Proposed sections:

1. Introduction
2. Related Work (three strategies, three levels)
3. Framework: Three Representation Levels
4. The ER Pipeline: Compare → Classify → Cluster
5. Three Fusion Strategies (A, B, C) and Normalization
6. Experiments (Exp 1–5)
7. Discussion: predictive diagnostics
8. Conclusion

Target: IEEE TKDE or VLDB Journal
Companion: R Journal software paper (`erbot.tex`)

Progress:

Task	Done?
Strategies A and B coded	✓
Restaurant, DBLP-ACM data	✓
Framework design (A/B/C)	✓
`er_snf()`	○
`er_normalize()`	○
`er_classify()`	○
`er_compare_paradigms()`	○
Experiments 1–5	○
Write paper	○

Summary

Contribution

First systematic comparison of three multi-view fusion strategies across three representation levels and four ER benchmarks (small to large scale). Key insight: choosing the right framing matters more than the algorithm — and the right framing is predictable from field-level diagnostics.

A — Late Clustering

- Fuse $\{S^{(k)}\} \to$ classify $\to$ cluster once
- MKL family; learned weights
- Best: complementary, compatible-scale fields

B — Early Clustering

- Classify per $S^{(k)} \to$ majority-vote consensus
- Parallelizable; robust to noisy fields
- Best: independent fields; large n

C — Hybrid

- SNF (diffusion) or multiplex community detection
- Best: complementary local structure; different relation types
- GNN bridges Level 3 and Level 1

Next: implement `er_snf()`, `er_classify()`, `er_normalize()`, then run Exp 1–5.